

Quantum Technologies Research Project

February Report

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4 Mar 26

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Project Goals & Metrics

Research Gap: No unified comparison of BB84 vs MDI-QKD across key metrics

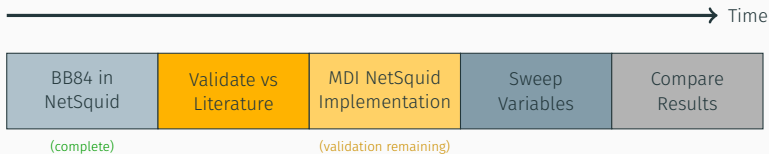
**Key Rate vs
Distance**
(varying networks)

**Implementation
Cost**
(economic & energetic)

Scalability
(user count)

- Analyse both protocols across varying network topologies
- Identify parameter regimes & topologies where each excels
- Inform practical QKD deployment decisions

Recap: Where We Were



Implementation stages:

1. Finalise BB84 implementation in NetSquid
2. Validate against known rate-distance curves from literature
3. Adjust BB84 → MDI-QKD with symmetric setup (+ validate)
4. Sweep distances, topologies and scale
5. Comparative analysis across metrics

Actions From Last I

From previous meeting:

- Review of EB cost modelling paper 
- What do I want to prove? 
- Relative protocol performance as modelling scales 
- Review of PLOB in reference to MDI 
- Literature Review - topic and begin drafting 
- **Preliminary Results** 

Vasileios K. - Comparative Cost Analysis of PM/EB QKD Networks

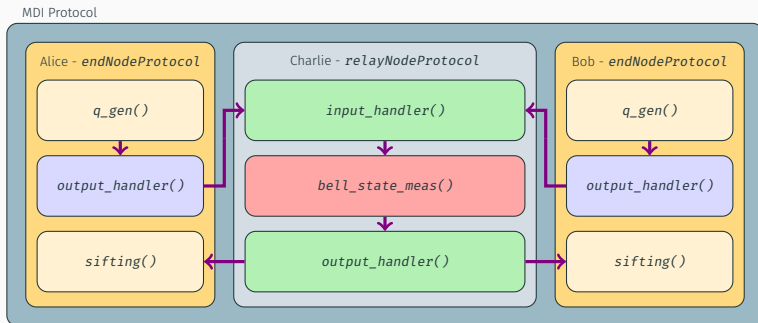
- **Modelling principle:** Fix one variable, optimise another (e.g. cost, key rate, respectively)
- Application of WDM to MDI - proof of principle demonstrated by Yan et al. 2025 (cited in Lit. Rev.)
- Integer Linear Programming toolkit - potentially applicable

Research Direction

- **Protocol-level** comparison
- Naturally lends itself to “**fairness of modelling**” and analysis of modelling level contributions - if time

NetSquid - Progress

- P2P simulators built for BB84 and MDI
- Scripting to enable:
 - Key metric tracking (QBER, rate, length)
 - Comparative analysis enabled
- Parametrisable by design - simple modelling integrations
- “CLI tool” for ease of use



NetSquid - Preliminary Results

- Key rates **obtainable**
- Variable distance, delay but no loss
- Idealised key rates in Mbps range
- Scripting setup - **parameter flexibility** + modelling extension
- MDI success rate 30-50% - protocol hangs sometimes, issue with asynchronous events *to be fixed*
- **Zero QBER** for both in idealised sims

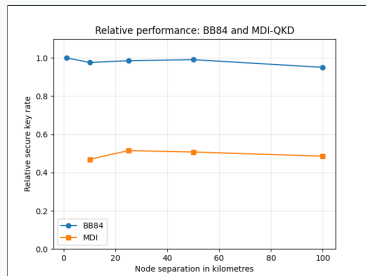


Figure 1: Key Rates for two protocols (idealised).
Photon count: 1024, Source frequency: 10^7 Hz, Photon speed: $0.8c$, Distance: variable

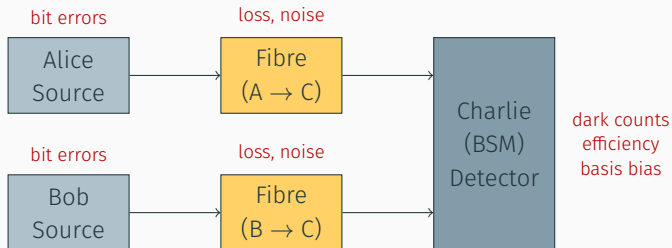
Next Stages - Component Modelling & Validation

Modelling to do:

- Fibre loss model
- Basic detector models (dark counts, inefficiencies)
- Random bit errors (source)

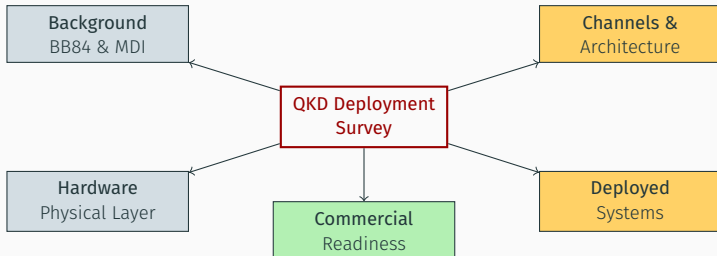
Validation to do:

- Reproduce key rates from parameters used by Lo et al. and other core papers (both protocols)



Literature Review: Title, Scope & Plan

- **Title:** Physical and Architectural Requirements for QKD Deployment:
A Survey of Implementation Platforms
- **Scope:**
 - *Background: BB84 & MDI* - introduce protocols
 - *Hardware and Physical Layer* - components of MDI
 - *Transmission Channels and Network Architecture* - MDI fibre, WDM and integration with classical channels
 - *Deployed Systems and Testbeds* - e.g. Q*Bird Port of Rotterdam, DARPA testnet
 - *Commercial Readiness and Outlook* - e.g. enterprise/national partnerships, remaining barriers, why comparative work needed



Completed

- NetSquid P2P simulators (v0.1) built
- Idealised comparison made

In Progress

- Parameter selection
- Validation benchmarking
- Modelling assumptions documentation
- Planning for scaling modelling
- Literature Review

Next immediate steps: Complete and submit Literature Review; validation of BB84 + MDI; hardware step-up modelling plan – one-pager – for P2P

Strategic decisions:

- Lit. Rev. - page count/technical content?
- Use of Vasileios' modelling philosophy?
- What constitutes sufficient simulation validation? Parameter matching/rate curve agreement?

Implementation priorities:

- Use of Vasileios' ILP models (given time & complexity)?